

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: OPL-YD-2301-03 Sheet 1 of 1
--	-------------------------------	---

OPL Title	Chain & Sprocket Lubrication and Tension (SCM440)	Discipline	Mechanical
Equipment	YD-2301 - POLYMER FLUID BED DRYER	Area / Unit	2300 - POWDER DRYING SYSTEM
Related Interlock	SEQ-5500 (SIL 2)	P&ID; Ref	TJC-LLD-PID-2301

Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E
------------------------	---	--------------------------------------	---------------------------------------	-----------------------

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **chain & sprocket lubrication and tension (scm440)** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. A dry or slack drive chain slips and drops the drum speed, tripping SSLL-2305 and stopping drying. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Weekly apply the specified chain oil along the full drive chain	Isolation & LOTO verified
2	Inspect the SCM440 sprocket teeth for hooking/wear	Parts match BOM / spec
3	Check chain tension - mid-span sag should be 10-15 mm	Torque / clearance to spec
4	Adjust the take-up if sag is excessive and re-lube	No leak / smooth operation
5	Record chain elongation; replace the chain at 3 % stretch	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- A dry or slack drive chain slips and drops the drum speed, tripping SSLL-2305 and stopping drying.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Friday, 10 April 2026